Lab Report: Text classification with neural networks

1 Introduction

In this report we compare several models on standard benchmarks. Each experiment was repeated 3 times with different random seeds. The dataset was split into 70 percent training data and the rest for testing.

2 Results

Figure 1 shows the test accuracy of Baseline by dataset.

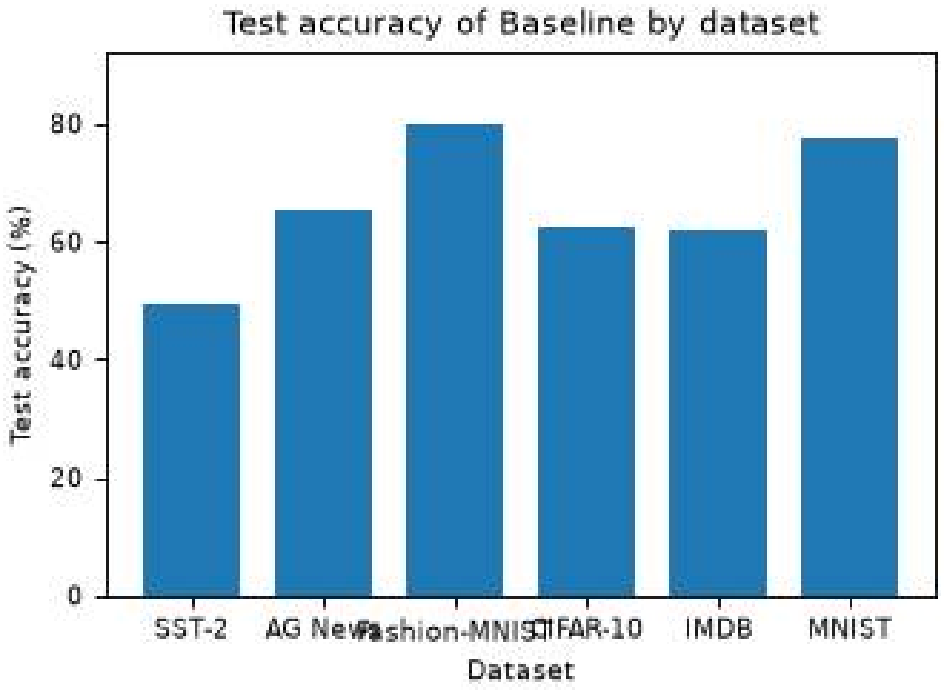


Figure 1: Test accuracy of Baseline by dataset for the models in this experiment.

Baseline scores 26.5 points higher on Fashion-MNIST than on CIFAR-10. Baseline reaches 62.2% test accuracy on MNIST.

Figure 2 shows the validation accuracy of ResNet-50 by dataset.

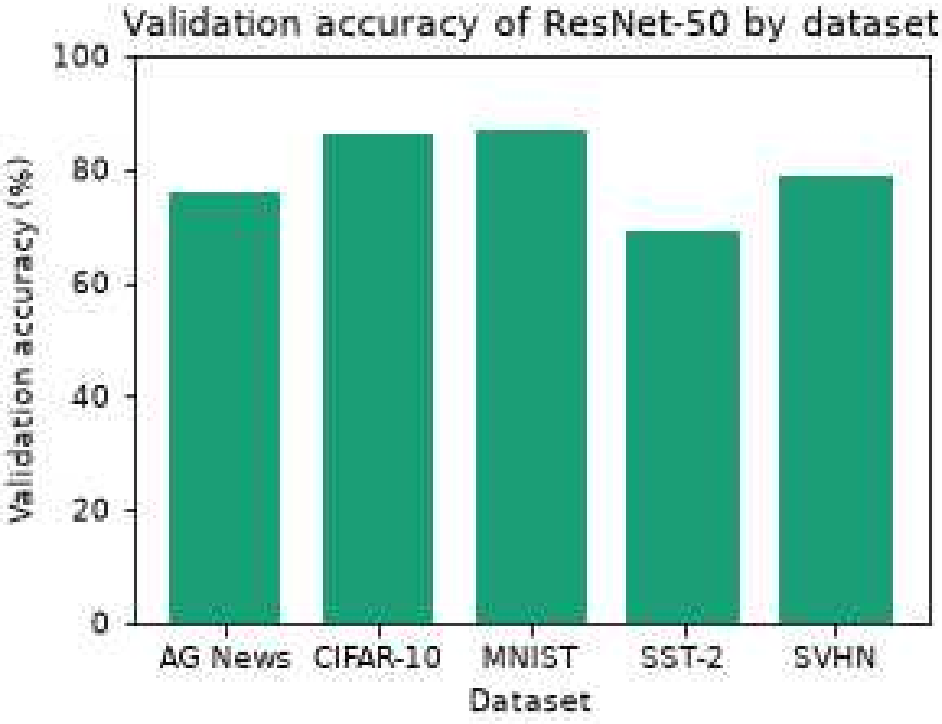


Figure 2: Validation accuracy of ResNet-50 by dataset for the models in this experiment.

As shown in Figure 2, ResNet-50 scores 2 points higher on SVHN than on AG News. ResNet-50 achieves 86% validation accuracy on CIFAR-10. ResNet-50 scores 2.0 points higher on AG News than on SST-2. As shown in Figure 2, ResNet-50 achieves 69.1% validation accuracy on SVHN.

3 Discussion

Logistic Regression reaches 72.7% accuracy on MNIST. We note that the training was stable for all models.